

Sharada Sahoo

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Summary

PhD in Physics (GPA 4.0) with over four years' experience in production environments, collaborating with engineers, physicists, and vendors to **develop particle detectors** for large-scale physics experiments. Extensive background in semiconductor and superconductor material fabrication, photon sensor characterization, and machine learning and data analysis using Python. Adaptable and results-driven, with a track record of contributing effectively across multiple projects to deliver impactful outcomes.

Education

Texas A & M University, College Station **2021 - 2025**
PhD candidate, Physics

CGPA - 4.0/ 4.0

- **Dissertation Title:** Dark matter searches using Ton scale CsI(Tl) detector and gram scale superconducting sensor based sapphire detector.
- **Relevant Coursework:** Quantum mechanics 1 & 2, Machine learning, Statistics, Electromagnetism 1 & 2.

National Institute of Science Education and Research, India **2016 - 2021**
BS - MS, Physics

CGPA - 8.0/ 10

- **Dissertation Title:** Rare event search analysis using Si based detectors
- **Relevant Coursework:** Quantum mechanics 1 and 2, Advanced electronics theory and lab, Quantum information, Quantum field theory 1 and 2, Condensed matter physics.

Experience

WIMP Dark Matter search with Superconducting Cryogenic Sensors **2021 - 2023**
Graduate Assistant

- Developed automated Python pipelines for data analysis, eliminating manual workload by (100%) - [Github](#).
- Performed SQUID tuning, superconducting transition edge sensor (TES) biasing using LabView maintaining workflow for the group.
- Conducted large scale data acquisition (DAQ) with CAEN digitizers, large set data storage in binary (ROOT) and handling the data structure with PyROOT and other python libraries.
- Delivered key experimental results, including achieving a lowest baseline resolution of 40 eV and producing a background energy spectrum after extensive rejection cuts, comparable to other world-leading experiments, presented at the EXCESS 2024 conference.
- Collaborated with the team in end-to-end experiment construction, from dilution refrigerator operation to assembling multi-layer detector shielding, including installing and repositioning heavy plastic veto systems.

BSM particle searches using radioisotopes with CsI (Tl) Detector **2023 - Ongoing**
Graduate Research Assistant

- Demonstrated the feasibility of CsI(Tl) technology by designing and constructing a ~100 kg CsI(Tl) detector instrumented with PMTs, paving the way for ton-scale detector development at LANL and ORNL; proposal currently under review.
- Worked with multiple radioisotopes like ^{22}Na , ^{57}Co , ^{60}Co , ^{133}Ba , ^{46}Sc , ^{137}Cs , ^{90}Sr , ^{55}Fe and ^{241}Am ranging from 0.1 uCi to 100 uCi activity to characterize the detector.
- Facilitated efficient detector development through effective coordination with lab engineers, supervisors, students, and vendors, ensuring smooth workflows from early-stage assembly to initial characterization.
- Integrated and operated diverse DAQ systems (oscilloscopes, picoscopes, CAEN digitizers, HV supplies) and completed a full experimental electronics workflow, from testing and characterizing sensors to repairing PMT bases, ensuring complete detector integration.
- Worked with multiple sensor types (PMTs, SiPMs, APDs) and scintillators (Bicron plastic, CsI, liquid scintillator, NaI), optimizing performance through material and configuration testing.
- Automated high-voltage control and data acquisition using the CAEN HV wrapper library, enabling fully unattended runs and collecting over 600 hours of continuous experimental data.
- Developed real-time data analysis pipelines in Python, automatically processing incoming data and generating live plots during experimental runs ([Github](#)).
- Performed advanced analysis for beyond-Standard-Model searches, applied statistical methods to extract results, set sensitivity projections for BSM particles like ALP, Dark photon, Dark Scalar and Milli-Charged particles, and demonstrated detector performance for funding proposals.

Low Cost strip PET Scanner development

Summer 2023 & Fall 2023

Graduate Mentor

- Demonstrated the operation of a single-strip plastic scintillator as the foundational element for an early PET scanner prototype at the Texas A&M Physics Festival 2023.
- Mentored undergraduate students in small-scale PET scanner development, providing hands-on training in scintillator preparation (polishing, wrapping), DAQ setup, and Python-based data analysis, enabling successful experiment completion.
- Presented experimental results at multiple conferences and at the Texas A&M Physics Festival 2024, showcasing the system's ability to accurately localize radioactive sources with a 1-inch standard error.

Josephson Junction Fabrication and Qubit Simulation

Summer 2024 & Fall 2024

Graduate Research Assistant

- Conducted literature review and collaborated with material engineers and PI to initiate the fabrication of Josephson junctions from scratch at Aggiefab, TAMU.
- Performed initial material deposition via sputtering, measured film thickness using AFM and SEM, and calibrated the sputtering process. Implemented wet etching with corrosive chemicals to refine etch parameters, documenting procedures for future use by the team.
- Completed the first junction layer and oxidation layer; third layer fabrication was halted due to lack of equipment (e-beam evaporation) and technical guidance. Gained extensive hands-on experience with fabrication tools and developed the ability to work independently while consulting experts.
- Performed qubit-related simulations in HFSS to study environmental coupling effects using a binary antenna design; work discontinued due to graduation.

Cryogenic Noise Analysis and Improvement Efforts

Summer 2025 - Ongoing

Graduate Research Assistant

- Implemented EMI/RF shielding and grounding strategies for noise mitigation, reducing interference and enabling the first-ever low-mass WIMP dark matter detector.
- Characterized thermal, Johnson, SQUID, and warm electronics noise to simulate cryogenic detector performance, providing data-driven guidance for improved detector design in collaboration with the UC Berkeley team.
- Operated a cryogenic dilution refrigerator to 8 mK, integrating wiring and optimizing hardware to improve TES stability and expedite detector characterization.

Machine learning Projects

Fall 2024 - Ongoing

Individual work

- Achieved top grade (A) in Machine Learning at Texas A&M, ranking in the top 10% of the class.
- Collaborated on a team project estimating meal calories from multimodal data (images + glucose monitor readings), achieving 34% error on unseen data using XGBoost and neural networks.
- Participated in Kaggle “March Mania 2025,” developing neural network models based on 40 years of game statistics to predict NCAA basketball winners for 2025.
- Built a probabilistic MCMC-based model for one-year stock price forecasting as a statistics class project; currently developing a Streamlit-based website to deploy the model.

Leadership

- Taught 200+ undergraduates as a Teaching Assistant, simplifying complex physics concepts through interactive problem-solving sessions.
- Mentored 12+ undergraduate and graduate students on photon sensor experiments and Python-based data analysis.
- Conducted a 10-day workshop at NISER, India, training 6 graduate students on TES sensor fabrication.
- Participated in public outreach through the TAMU Physics Festival each year since 2022, demonstrating experimental physics concepts to thousands of diverse audience members.

Conferences and Seminar

- Texas APS 2023, APS 2024, APS 2025 – Delivered talks on axion searches and photon detector development.
- NISER, India (2023) – Seminar on superconducting TES sensor fabrication techniques.
- UT Austin (2025) – Seminar on axion and axion-like particle searches using cryogenic and room-temperature detectors.
- CIPANP 2025, University of Wisconsin–Madison – Presented dark matter search results from Dr. Mahapatra’s group using cryogenic and room-temperature detectors.
- UC Berkeley (2025)– Presented axion dark matter searches with CsI(Tl) detectors and discussed future research directions.

Publications

- Development of a large-mass, low-threshold detector system with simultaneous measurements of athermal phonons and scintillation light. (arXiv:2212.04342)
- Reactor-based Search for Axion-Like Particles using CsI(Tl) Detector (arXiv:2407.14704)

- MINER Reactor Based Search for Axion-Like Particles Using Sapphire (Al_2O_3) Detectors. (arXiv:2504.20960)
- A GEANT4-Based Simulation of Directional Neutron Detectors Using Liquid Scintillators and Boron Carbide Moderators. (arXiv:2507.14438)
- Search for Dark Matter via Invisible Decay in ^{46}Sc Nuclear Gamma Cascades Using a CsI (TI) Detector. (arXiv:2510.23001)

Skills

Programming & Software Development

- Languages: Python, C/C++
- Libraries & Frameworks: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Keras, PyTorch, TensorFlow
- Simulation & Design Software: HFSS, K-Layout
- Machine Learning model development, API development for software
- Text & Documentation: LaTeX, Word, Microsoft Office Suite, Google slides.
- Web Development: Streamlit, Render
- AI softwares: ChatGPT, Gemini, Github Copilot

Device Fabrication & Characterization

- Thin Film Deposition: DC/RF Sputtering
- Lithography Techniques: Mask & Maskless Aligner Lithography, Electron Beam Lithography
- Imaging & Characterization: Optical Microscope, Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM)
- Surface Profiling & Etching: DekTak, Wafer Dicing, Wet Etching

Experimental & Instrumentation Skills

- Electronic & Circuit Design: Advanced electronic circuit design, amplifiers, digitizers, HV/LV power sources
- Photon Sensors: PMTs, SiPMs, APDs, Silicon Photodiodes
- Detector Characterization & Data Acquisition: Performance testing and data collection for detectors in both low (~ 8 mK) and room temperature.
- Cryogenic Systems: 10 K Cryostat, Dilution Refrigerator
- Semiconductor Devices: Silicon (Si), Germanium (Ge)
- Scintillators: CsI, Sapphire
- Oscilloscopes: Oscilloscope, Picoscope

Relevant Courses:

- Physics : Solid State Physics, Electromagnetism I & II, Quantum Mechanics I & II, Quantum Field Theory I & II, Quantum Information, Condensed Matter Physics.
- Mathematics & Statistics: Machine Learning, Statistics
- Experimental & Applied Physics: Thermodynamics, Optics, Advanced Electronic Laboratory

Soft Skills

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| <ul style="list-style-type: none"> ● Research management ● Technical Writing & Documentation ● Conference Talks & Publications ● Project Coordination ● Proposal & Report Writing | <ul style="list-style-type: none"> ● Public Engagement & Outreach ● Academic tutoring ● Mentorship & Training ● Scientific Communication |
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Awards

- KVPY Fellowship – Achieved an All India Rank of 9 in the prestigious Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship.
- NGPE – Ranked among the top 1% in the National Graduate Physics Examination (NGPE).
- NET – Secured an All India Rank of 95 in the National Eligibility Test (NET), India.
- JEST – Successfully cleared the Joint Entrance Screening Test (JEST).
- IISc Admission – Selected for graduate studies at the Indian Institute of Science (IISc), India.
- Olympiad – Qualified for the State Junior Mathematics Olympiad and participated in summer training camps.

Extra - Curricular

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| <ul style="list-style-type: none">● Running – Participated in two 5K running events and 2 half marathon events.● Cricket – Played cricket tournaments for Texas A&M. | <ul style="list-style-type: none">● Badminton, Table Tennis, Swimming.● Mountain climbing, music, and group activities |
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